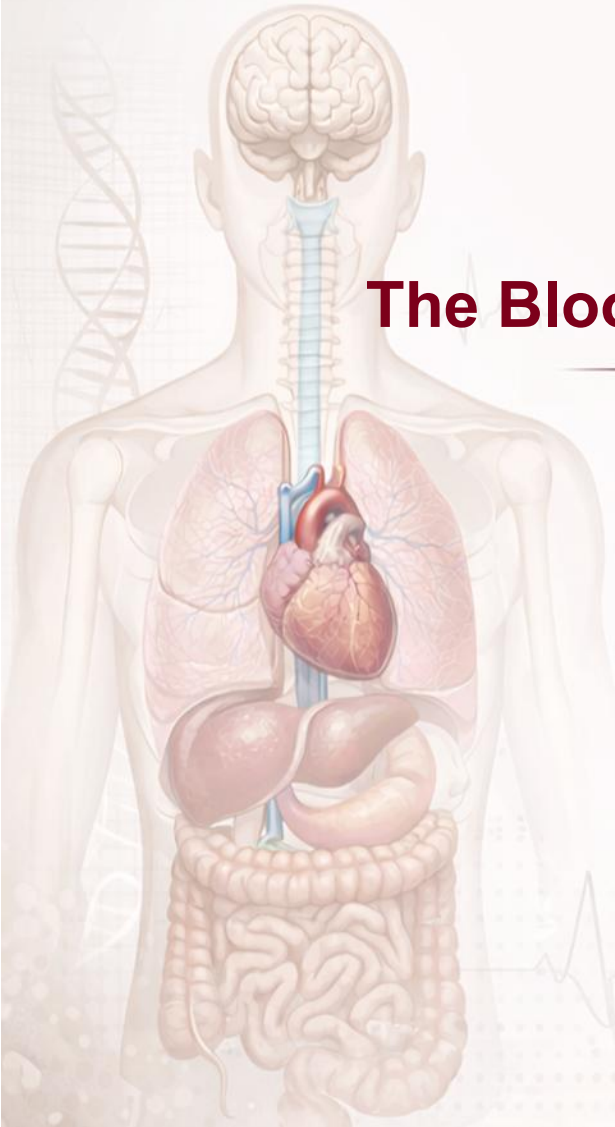


Chapter 10.
The Blood Vessels and Blood Pressure

KOREA UNIVERSITY
1905

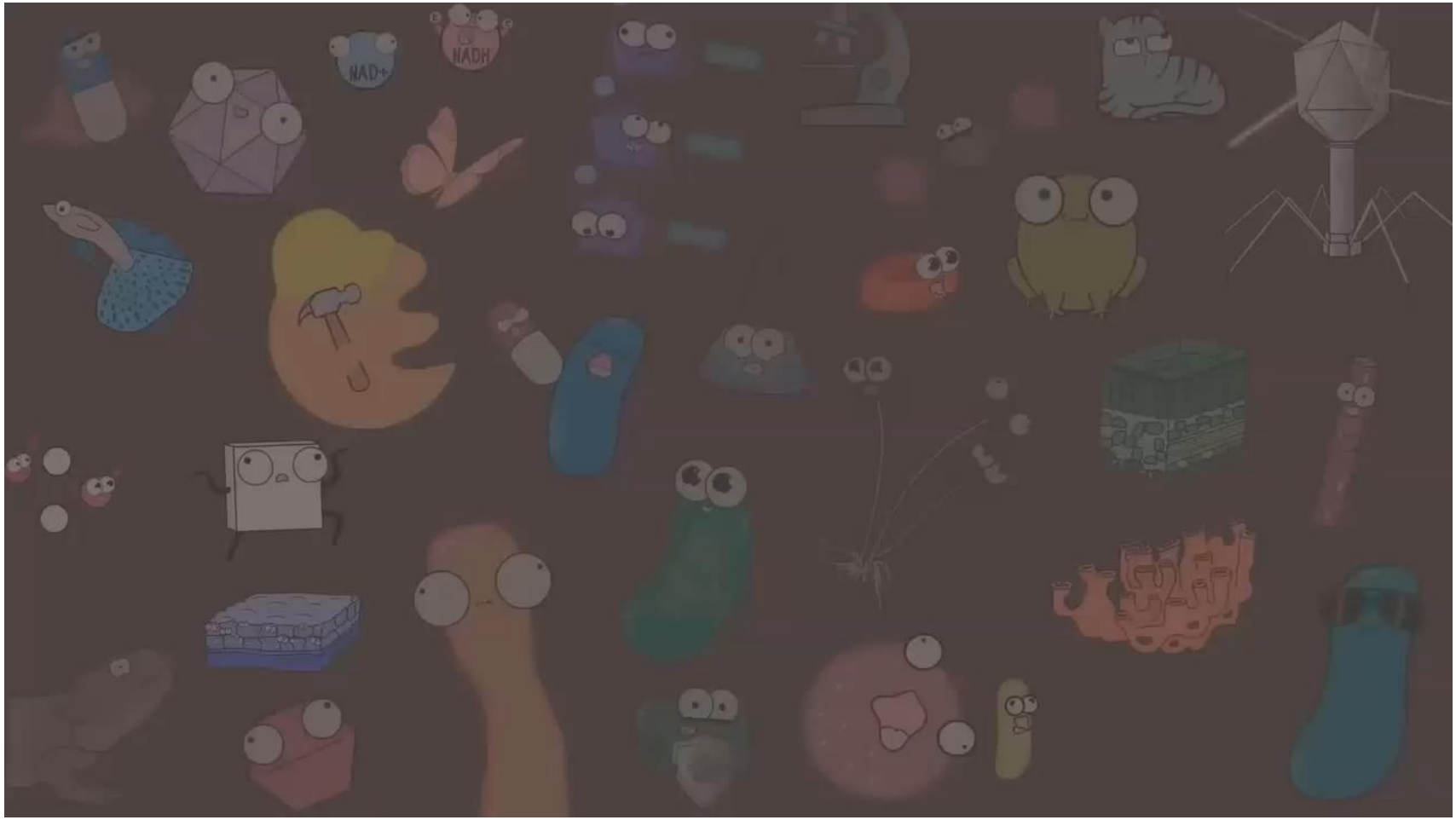
A scanning electron micrograph of blood vessels supplying the small intestine



© Susumu Nishinaga/Science Source

10.1 Patterns and Physics of Blood Flow

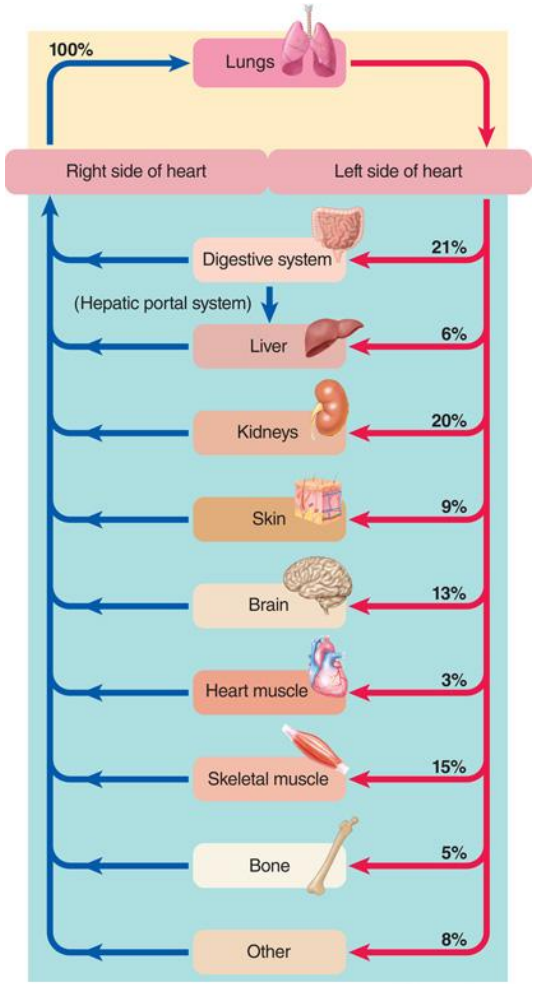
Circulatory system and pathway of blood through heart



https://www.youtube.com/watch?v=_vZ0lefPg_0

Blood flow

- Blood is transported to all parts of the body through a system of vessels
  - Most body cells are not in direct contact with the external environment
  - Blood brings fresh supplies to the vicinity of all cells while removing their wastes + achieve long-distance signaling molecule exchange
- Blood vessels: **passageways** where blood is directed and distributed from the heart to all body parts and finally returned to the heart
  - Parallel arrangement of blood vessels ensure that all organs receive blood of same composition
  - Highly elastic **arteries**
  - **Arterioles**, composed of 1-2 layers of smooth muscle cells, regulate blood pressure
  - Smallest blood vessels (**capillaries**) are designed for rapid material exchange with the surrounding tissue
  - Highly expandible **veins** serve as a blood reservoir



Circulatory system

- Reconditioning organs receive blood flow in the excess of their needs
  - So that they can adjust the extra blood to achieve body homeostasis and withstand temporary reductions in blood flow better
  - Kidney (electrolyte, waste), digestive tract (nutrients), liver (metabolism, detox), skin (body temperature), endocrine (hormone)
  - Blood to the other organs (brain, heart, skeletal muscle) is solely for filling these organs' metabolic needs
  - Adjustment of CO distribution according to the body's current needs
- Blood flow through a vessel depends on
  - Flow rate (volume of blood passing through / time)

$$F = \Delta P/R$$

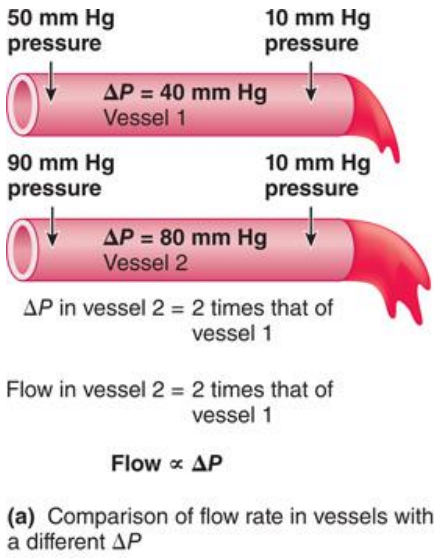
F = flow rate of blood through a vessel

ΔP = pressure gradient

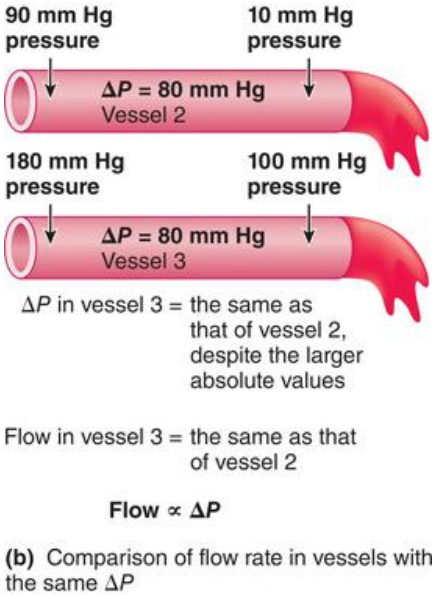
R = resistance of blood vessels

Factors in blood flow

- **Pressure gradient**: The difference in pressure between the beginning and end of a vessel
  - Heart contraction imparts the pressure to the blood (main driving force for blood flow)



(a) Comparison of flow rate in vessels with a different ΔP



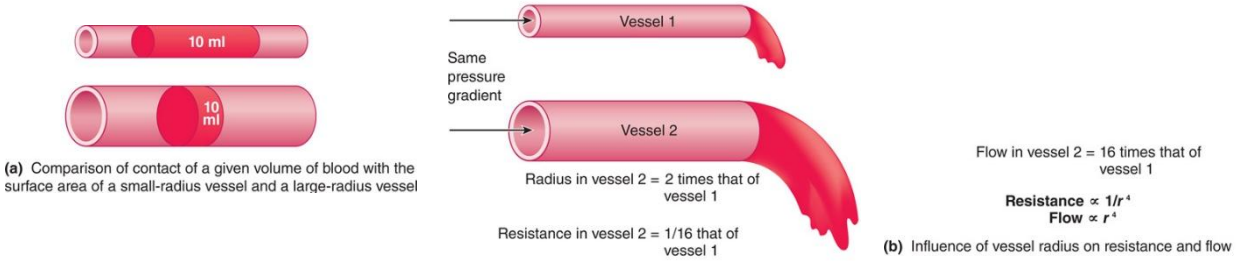
(b) Comparison of flow rate in vessels with the same ΔP

Factors in blood flow

- **Resistance**: Hindrance to blood flow through a vessel
  - Frictional losses (resistance) in the blood vessel decrease the pressure
  - When resistance increases, the pressure gradient must increase (=heart must work harder) proportionally to maintain the same blood flow rate
  - Resistance to blood flow is directly proportional to (1) viscosity of the blood, (2) blood vessel length, and (3) inversely proportional to vessel radius

    $R \propto \eta L/r^4$

    R = resistance / η = viscosity / L = vessel length / r = vessel radius
  - Viscosity (blood composition) and vessel length is not a variable factor
  - Major determinant of resistance to flow is the vessel's radius
  - Blood passes rapidly through a large vessel than through a small vessel

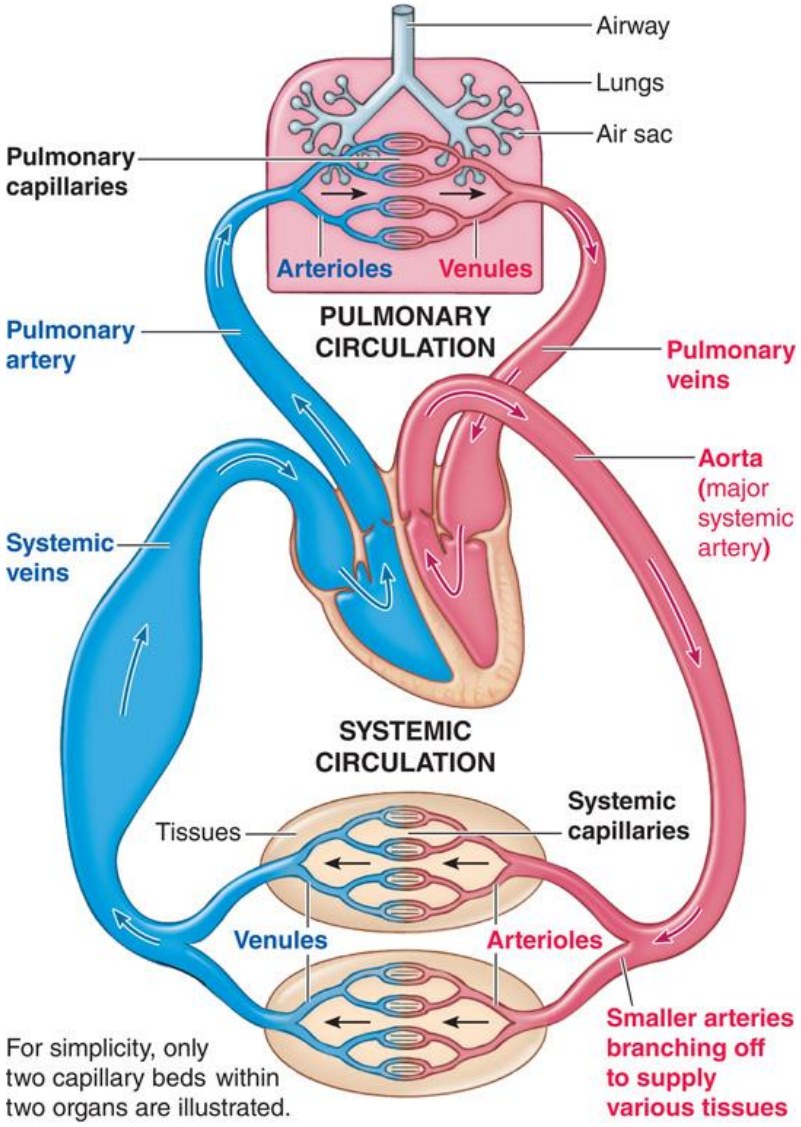


- Poiseuille's law (푸아죄유 법칙)
  - Integrates factors affecting flow rate through a vessel
  - Q = flow rate

$$Q = \frac{\Delta P \pi r^4}{8\eta L}$$

The vascular tree

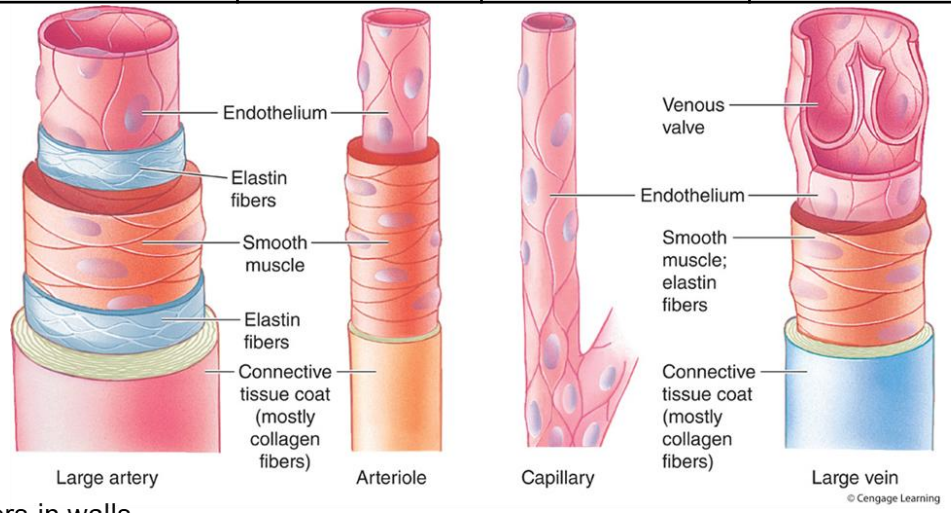
- **The vascular tree** is a hierarchical, branching network of blood vessels—arteries, capillaries, and veins—that distributes O_2 and nutrients to tissues and returns waste, acting as a "closed system“
  - Arteries transport blood from the heart to organs
  - Arterioles (세동맥) control the amount of blood that flows through each organ
  - Capillaries are vessels where materials are exchanged between blood and cells
  - Capillaries rejoin to form venules (세정맥)
  - Veins return blood to the heart
  - Microcirculation = arterioles, capillaries, and venules that are visible only through a microscope

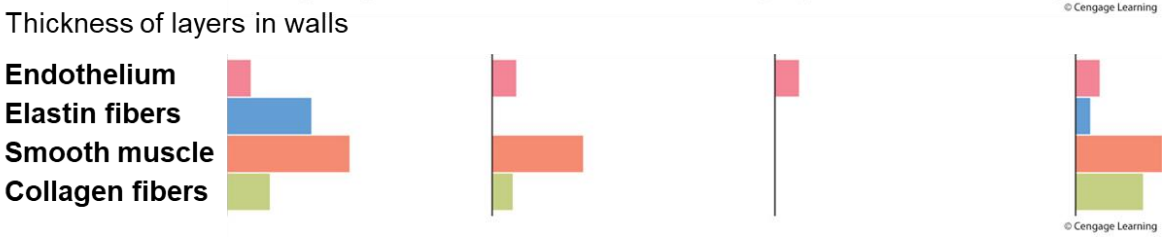


10.2 Arteries

Consecutive segments of vascular tree

	Arteries	Arterioles	Capillaries	Veins
Number	Several hundred	500000	10 billion	Several hundred
Special features	Thick, highly elastic walls	Highly muscular, highly innervated with SNS	Vert thin walls, large total cross-sectional area	Thinner than arteries, highly expandible walls
Functions	Passageway from heart to organs, pressure reservoir	Primary resistance vessels	Site of exchange, determine ECF distribution between plasma and interstitial fluid	Passageway from organs to heart, blood reservoir

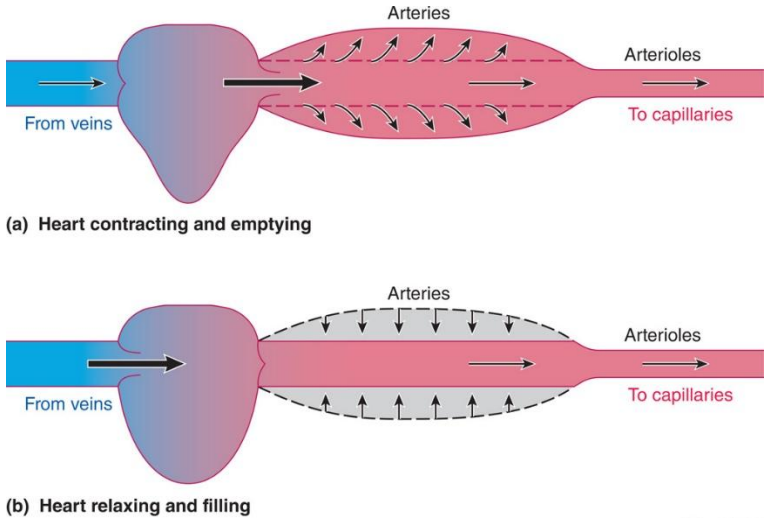




Arteries

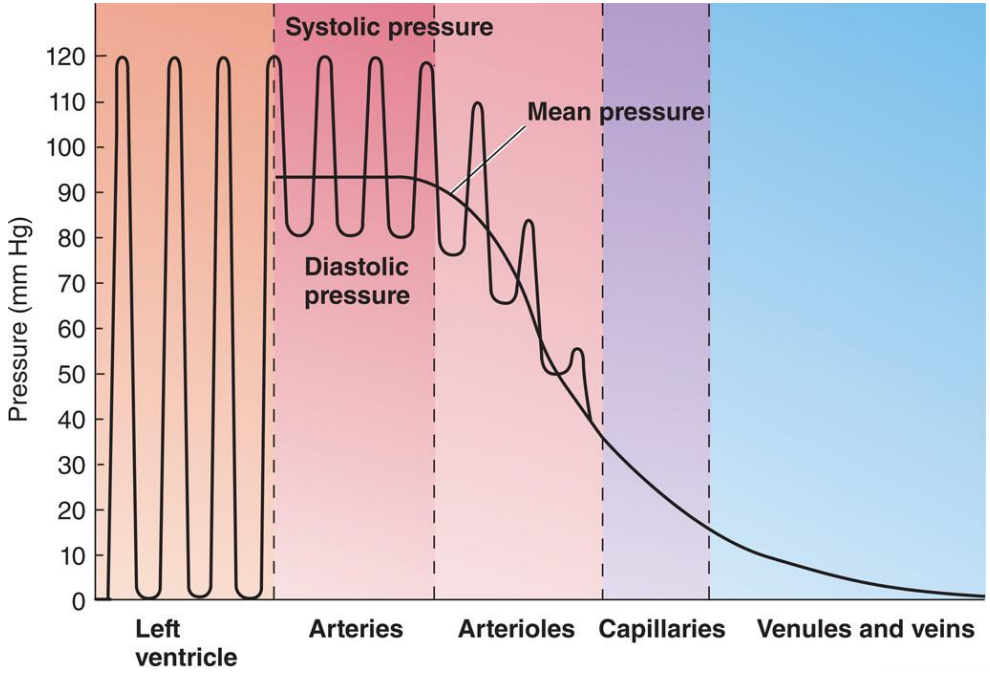
- Arteries serve as (1) **rapid-transit passageways** to the organs and as (2) a **pressure reservoir**
  - Provide the driving force for blood when the heart is relaxing (=no blood is pumped out from the heart)
  - However, blood flow is still continuous through the capillaries and this continuous flow is provided by **elastic recoil** of artery walls

- Artery vessels are lined with endothelium (endothelial lining)
- A thick wall made up of smooth muscle and connective tissue surrounds the endothelium
  - collagen fibers: provide tensile strength
  - elastin fibers: provide elasticity
- Elastic recoil during diastole exerts pressure on the blood in large arteries



Mean arterial pressure (MAP)

- Mean arterial pressure is the main driving force for blood flow
- MAP = average pressure driving blood forward into the tissues throughout the cardiac cycle
- At resting heart rate, about two thirds of the cardiac cycle is spent in diastole -> MAP is closer to diastolic pressure



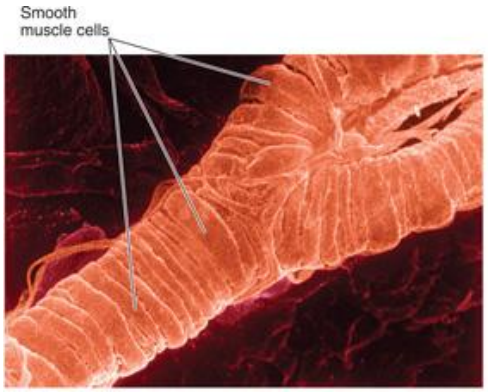
10.3 Arterioles

Arterioles

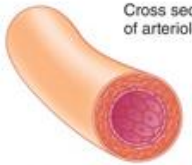
- When an artery reaches the organ it is supplying, arteries branch into numerous arterioles within the organ
- Arterioles are the **major resistance vessels**, not capillaries
  - Their radius is small enough to offer significant resistance to flow
  - High degree of arteriolar resistance causes a marked drop of pressure
  - This decline in pressure is **helpful for establishment of the pressure gradient** triggering the blood flow from the heart to the various organs
  - Arteriole radius can be adjusted to accomplish two functions
    - Variably distribute the CO among the organs, depending on the body's momentary needs
    - To help regulate arterial blood pressure

- Although elastic connective tissues of arteriolar walls are thin, they do have a thick layer of **smooth muscle richly innervated by SNS**
  - Vasoconstriction: narrowing of vessels
  - Vasodilation: enlargement of vessels as a result of smooth muscle relaxing

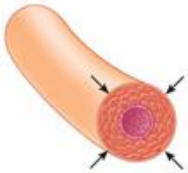
Arterioles



(a) Scanning electron micrograph of an arteriole showing how the smooth muscle cells run circularly around the vessel wall

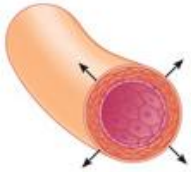


(b) Normal arteriolar tone



Caused by:
- ↑ Myogenic activity
- ↑ Oxygen (O_2)
- ↓ Carbon dioxide (CO_2) and other metabolites
- ↑ Endothelin
- ↑ Sympathetic stimulation
- Vasopressin; angiotensin II
- Cold

(c) Vasoconstriction (increased contraction of circular smooth muscle in the arteriolar wall, which leads to increased resistance and decreased flow through the vessel)

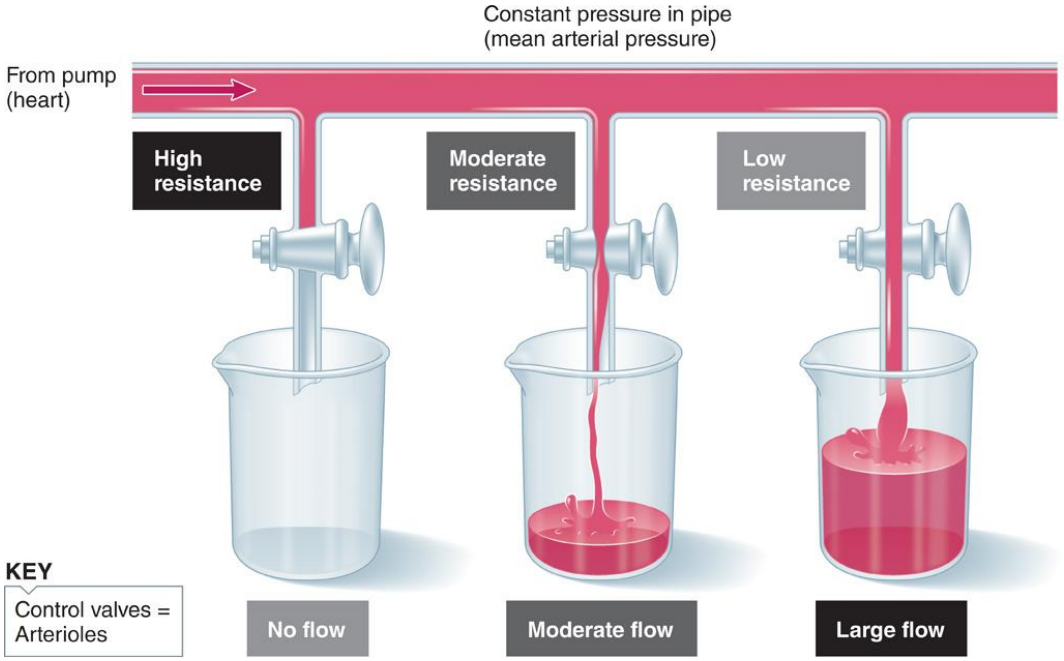


Caused by:
- ↓ Myogenic activity
- ↓ O_2
- ↑ CO_2 and other metabolites
- ↑ Nitric oxide
- ↓ Sympathetic stimulation
- Histamine release
- Heat

(d) Vasodilation (decreased contraction of circular smooth muscle in the arteriolar wall, which leads to decreased resistance and increased flow through the vessel)

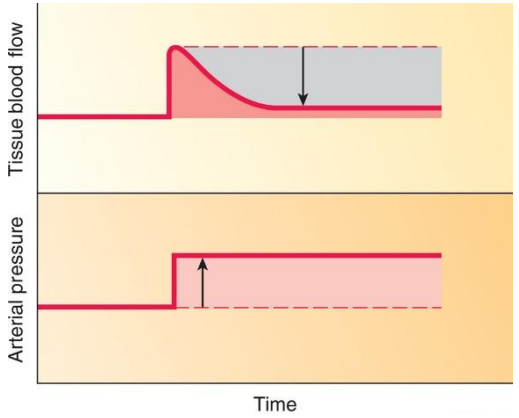
Local (intrinsic) control of arteriole radius

- Local control of arteriolar radius is important for determining the **distribution of CO**
  - The fraction of total CO delivered to each organ is controlled by the blood demands
  - Differences in flow to organs are determined by differences in **vascularization** and differences in **resistance** offered by arterioles supplying each organ



Local (intrinsic) control of arteriole radius

- Intrinsic control of arteriolar radius: changes within an organ that adjust blood flow through the organ by affecting the smooth muscle of its arterioles
- Local influences can be chemical or physical
  - Chemical influence on radius
    - Local metabolic changes
      - Decreased O_2
      - Increased CO_2, acid, K^+, osmolarity
      - Adenosine release (for cardiac muscle fibers)
    - Histamine released by mast cells during inflammation and injury
  - Physical influences
    - Myogenic response: perfusion pressure → stretch of the arteriolar wall → vasoconstriction → organ blood flow remains constant (blood pressure auto-regulation)
    - The extent of shear stress: endothelial cells release NO under shear stress
    - Local application of heat or cold



Extrinsic control of arteriole radius

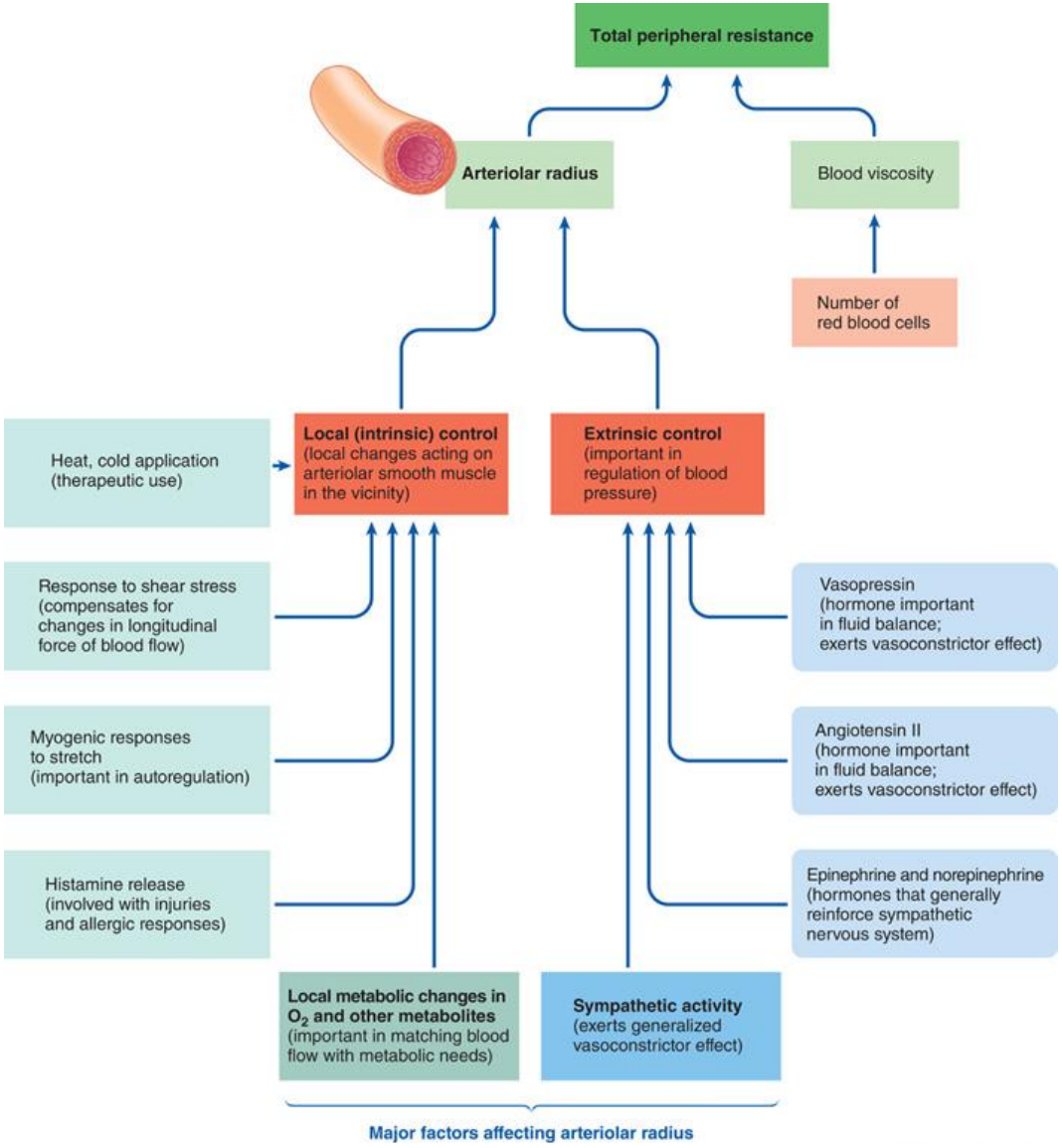
- Extrinsic control of arteriolar radius is important in regulating **blood pressure**

- Sympathetic nervous system releases norepinephrine to activate α-adrenergic receptor
  - induces vasoconstriction (smooth muscle contraction)
  - increases **total peripheral resistance (TPR)**
  - TPR: total resistance offered by all the systemic peripheral vessels
  - MAP = CO X TPR
  - increases blood pressure
- No parasympathetic innervation to arterioles
  - Vasodilation is induced by decreased sympathetic activity and local NO (nitric oxide) secreted from endothelial cells

The cardiovascular control center and hormones

- Cardiovascular control center in the medulla of brain stem is a center for blood pressure regulation
  - Sympathetic neural output to regulate HR, cardiac contractility, and arteriolar constriction
- Several hormones also extrinsically control arteriolar radius
  - **Epinephrine and norepinephrine** released by adrenal medulla reinforce sympathetic nervous system
  - **Vasopressin** maintains water balance by regulating the water amount for reabsorption (posterior pituitary gland, 뇌하수체 후엽)
  - **Angiotensin II** promotes the conservation of salt and water (capillary endothelium in the lung)
  - Both vasopressin and angiotensin II are also vasoconstrictors

Factors affecting total peripheral resistance

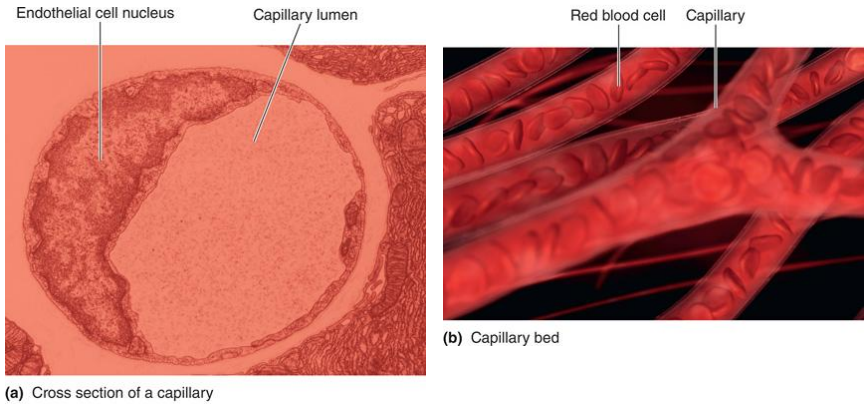


10.4 Capillaries

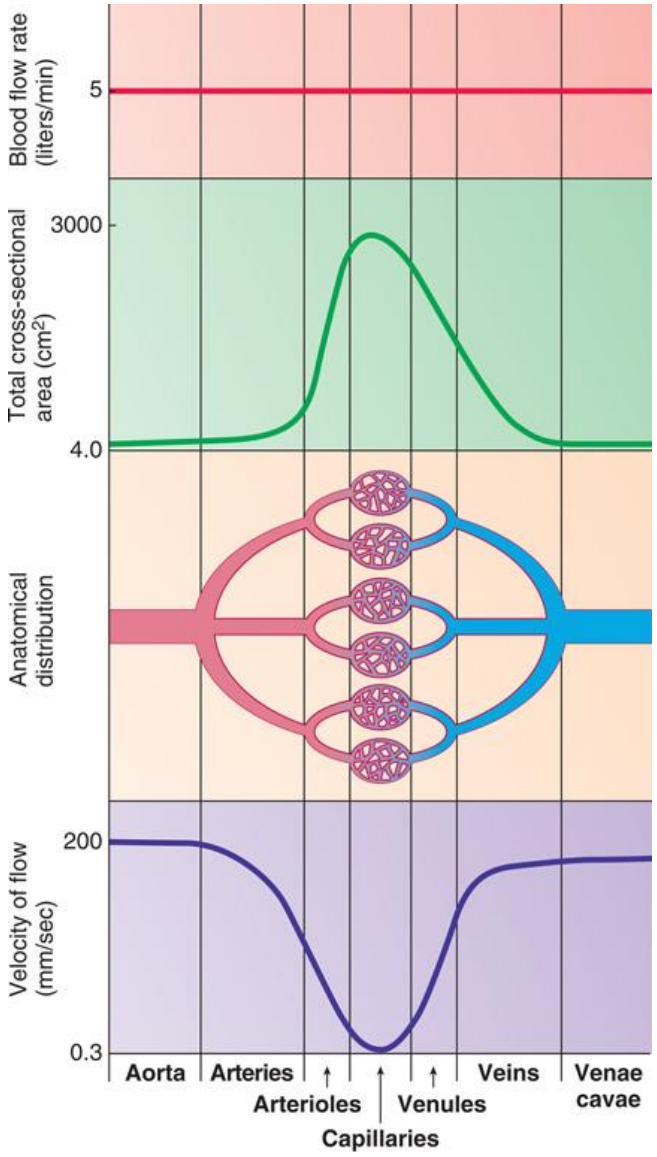
Capillaries

- Capillaries are sites for exchange of materials between blood and tissue cells, bringing blood to every cell
- Many substances diffuse across capillary walls
  - Except for BBB, O_2 and CO_2 diffuse, while glucose via GLUT1
- Diffusion across the capillaries is ideally suited
  - Diffusing molecules have only a short distance to travel between blood and surrounding cells (very thin, single layer)
  - Each capillary is so narrow that red blood cells have to squeeze through
  - Scarcely any cell is farther than 0.1 mm from a capillary
  - The number of capillaries and total surface area are incredibly high
  - Blood flows slowly than elsewhere

Diffusion distance
Surface area
Time



Capillaries



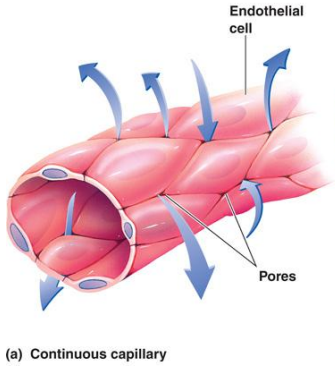
The **blood flow rate is identical** through all levels of circulatory system, equal to CO.

The **velocity of flow varies** and is inversely proportional to the total cross-sectional area.

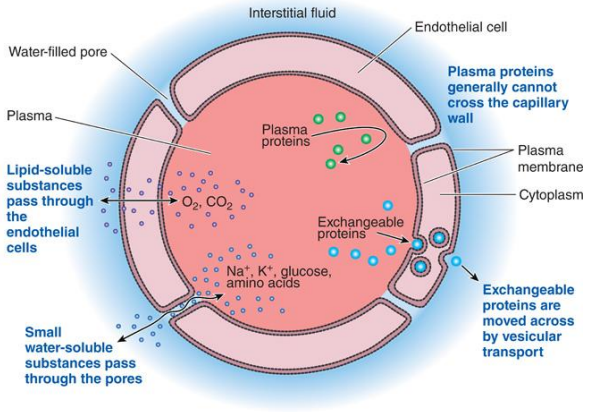
Capillaries are place for O_2 supply, nutrient supply, CO_2 removal, waste removal to organs.

Capillaries

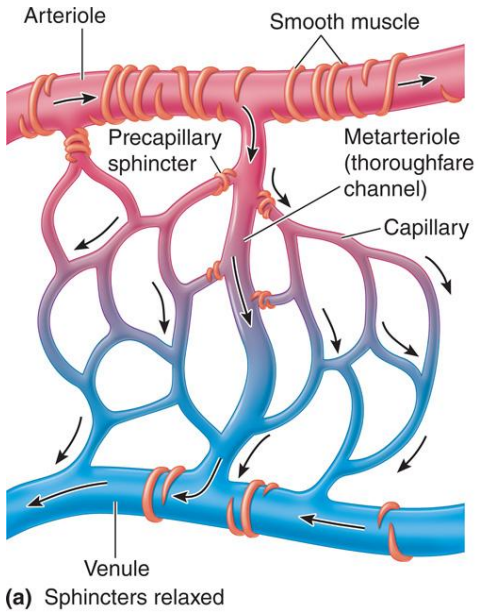
- Water-filled capillary **pores** between endothelial cells permit passage of small, water-soluble substances
  - Pores: narrow, water-filled clefts
- Many capillaries are not open under resting conditions
  - More capillaries are opened as tissue metabolic activity increases



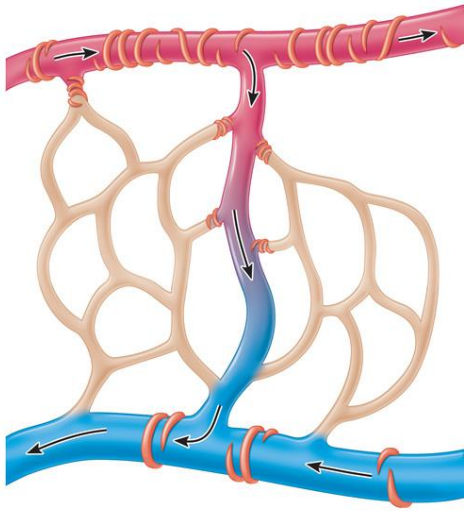
(a) Continuous capillary



(b) Transport across a continuous capillary wall



(a) Sphincters relaxed



(b) Sphincters contracted

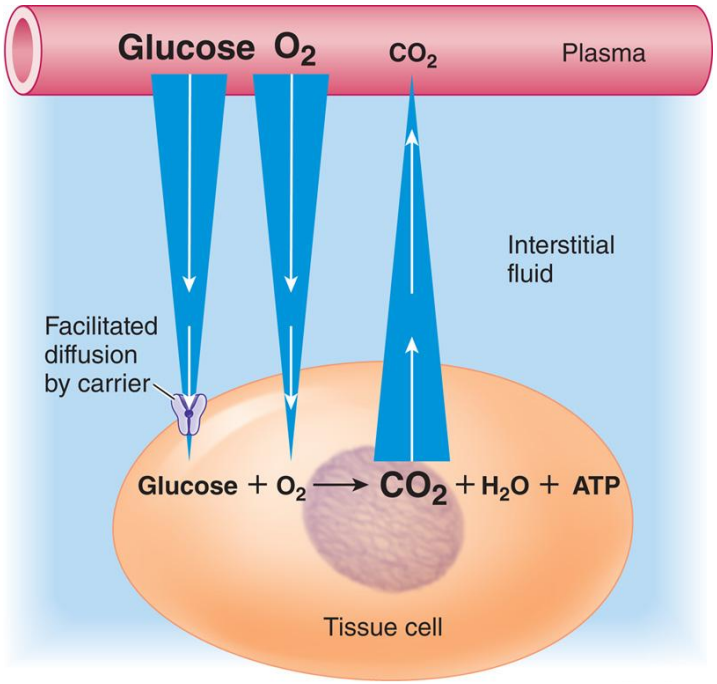
Precapillary sphincter: tiny ring of smooth muscle located at the capillary branch junction, acting as a valve to regulate blood flow into the capillary bed

Metarteriole: short microvessel that connects arterioles to capillaries

Interstitial fluid and diffusion

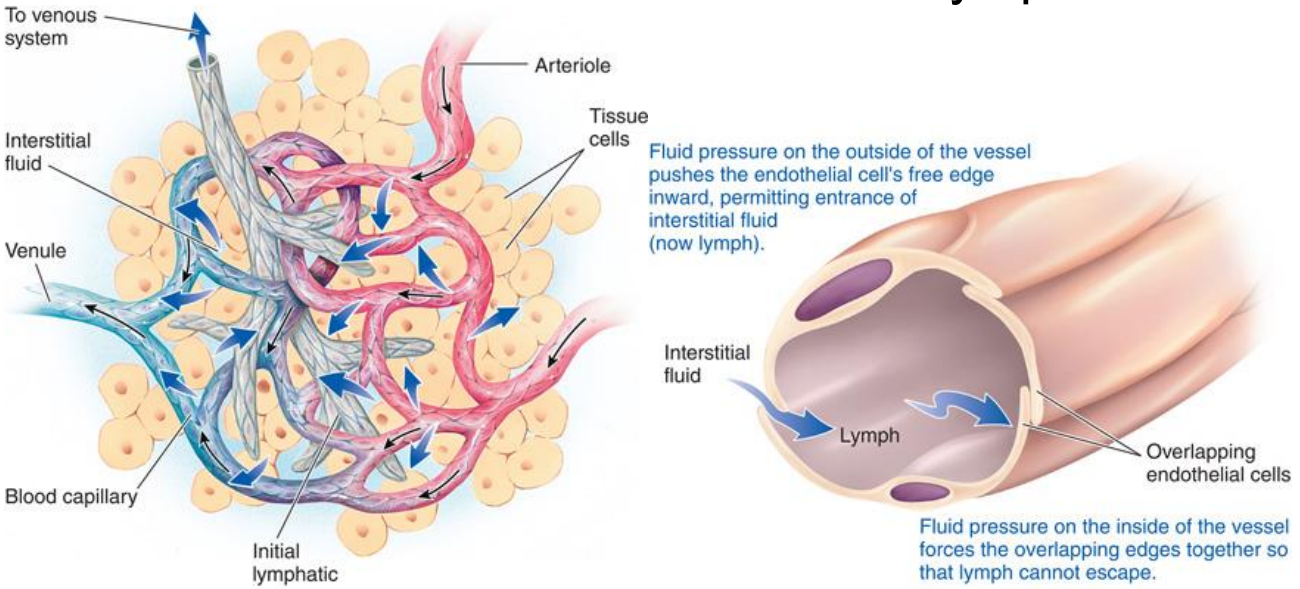
- Interstitial fluids are the true internal environment in immediate contact with the cells
- ECF = 80% interstitial fluid + 20% plasma
- Cells exchange materials directly with interstitial fluid, as a passive intermediary between blood and cells
- The exchange between interstitial fluid and cell is either passive or active, while exchange between the plasma and interstitial fluid is **mainly passive**

- Diffusion across capillary walls is important for solute exchange
- Extent of exchanges for each solute is independently determined by the magnitude of its concentration gradient between blood and surrounding cells



The lymphatic system

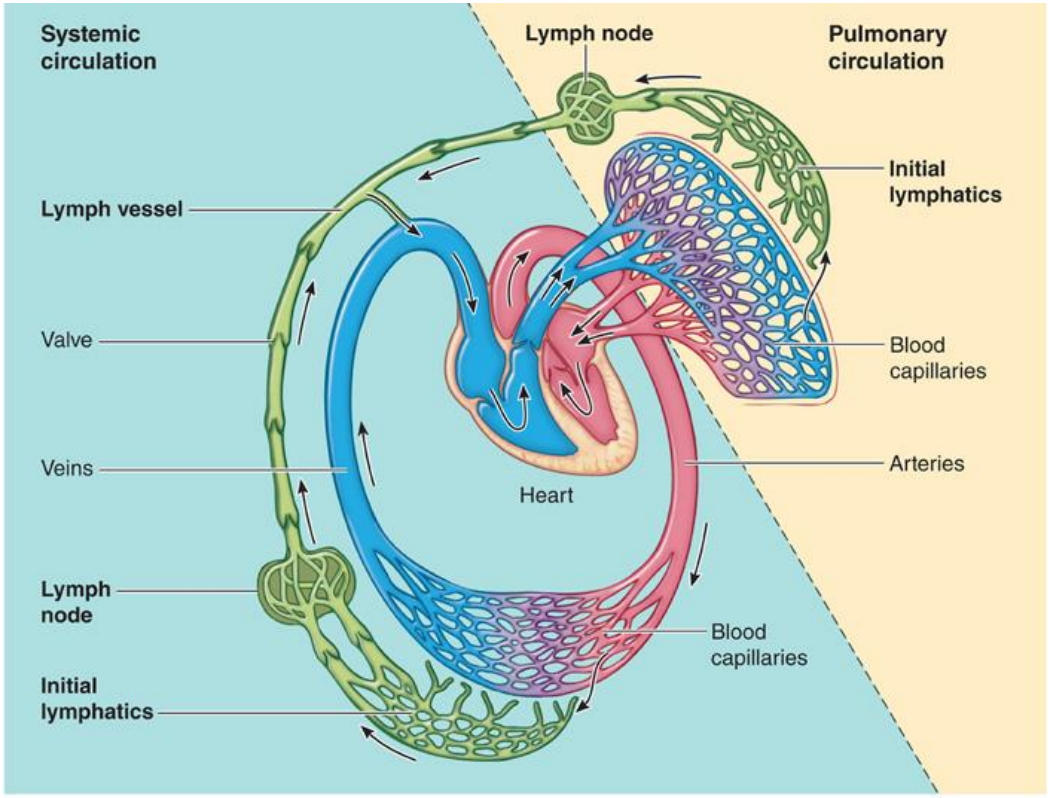
- Slightly more fluid is filtered out of the capillaries into the interstitial fluid than is reabsorbed back into the plasma
- The extra fluid filtered out is picked up by the lymphatic system
- This one-way route provides an accessory route for return of interstitial fluid to the blood
- **Initial lymphatics**: highly permeable smallest vessels of lymphatic system absorbing interstitial fluid, macromolecules, and immune cells
- **Lymphs:** the interstitial fluids that entered into a lymphatic vessel



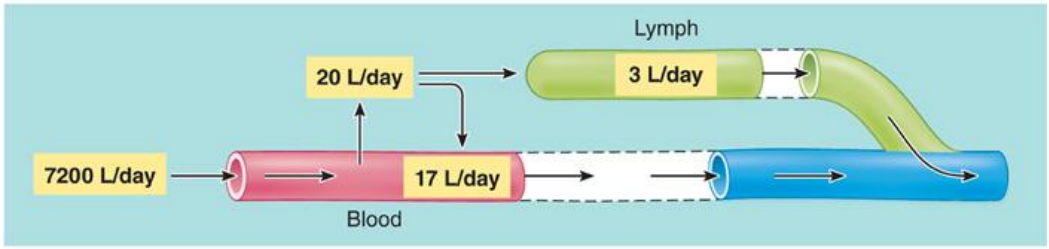
(a) Relationship between initial lymphatics and blood capillaries

(b) Arrangement of endothelial cells in an initial lymphatic

The lymphatic system



(a) Relationship of lymphatic system to circulatory system



(b) Comparison of blood flow and lymph flow per day

Structure of lymphatic system
Initial lymphatics
Lymph vessel: initial lymphatics converge to form larger lymph vessels
Lymph node: act as filters for the lymphatic system and a vital part of the immune system

Functions of lymphatic system
Return of excess filtered fluid
Defense against disease
- Passage of lymph through lymph nodes kills pathogens using phagocytes inside

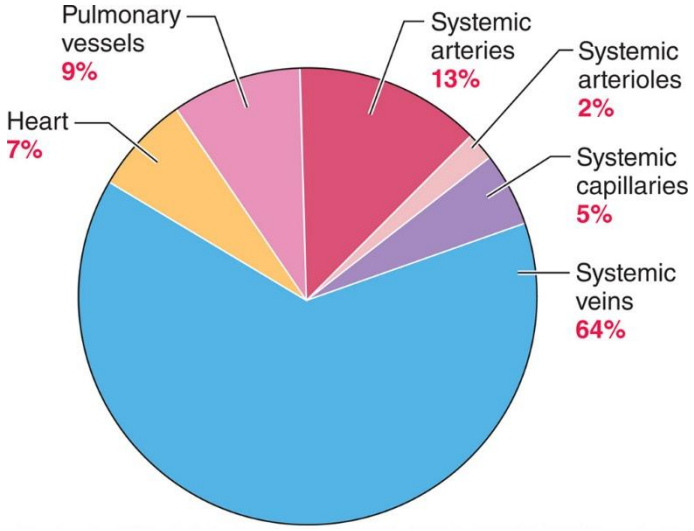
Transport of absorbed fat from the GI tract
Return of filtered proteins

10.5 Veins

Veins

- Venules communicate chemically with nearby arterioles
  - Vital to matching capillary inflow and outflow within an organ

- Veins serve as a blood reservoir and as passageways back to the heart
  - When demands for blood are low, veins can **store extra blood** in reserve because of their passive **distensibility**



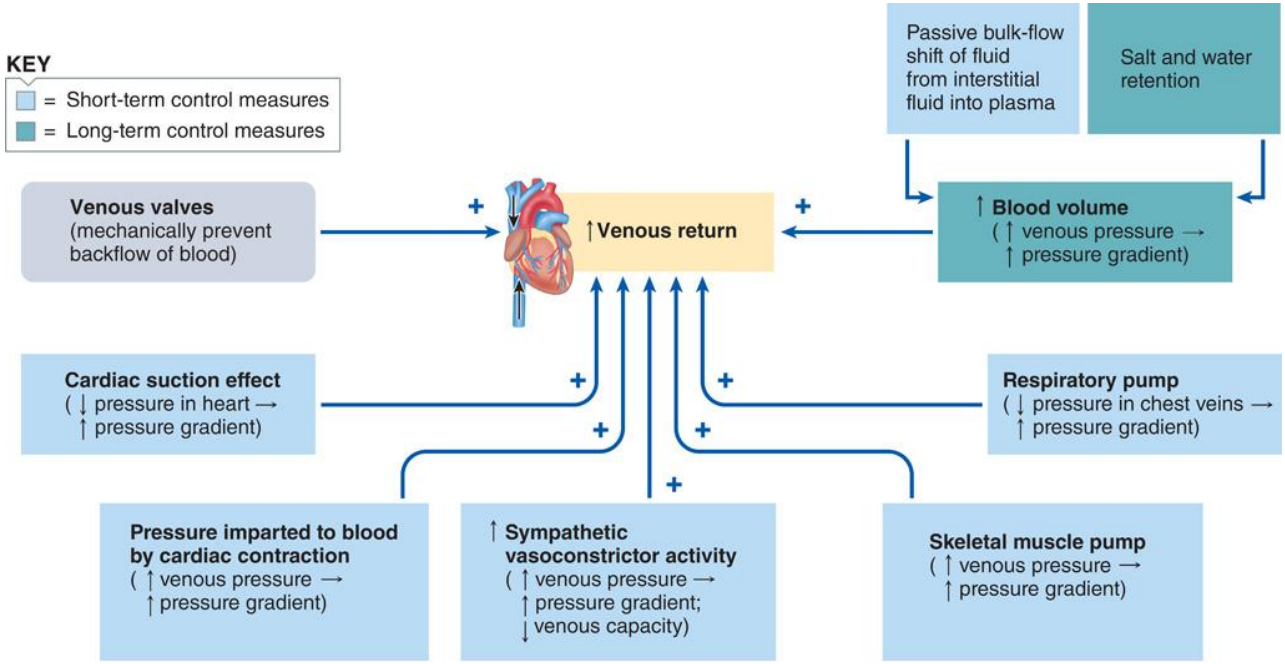
Venous return (정맥환류)

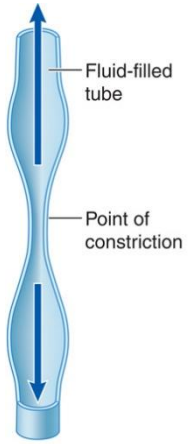
- Venous capacity (the blood volume that veins can contain) depends on the distensibility of the vein walls
- Increase of venous capacity reduces the amount of blood returning to the heart (decreased effective circulating blood volume)

- **Venous return**: the volume of blood per minute entering each atrium from the veins
  - Determines stroke volume and thereby cardiac output
  - Although the blood entering the venous system is only 17 mm Hg, atrial pressure is 0 mm Hg -> pressure gradient (**intrinsic factor**)
  - Five additional extrinsic factors increasing venous return

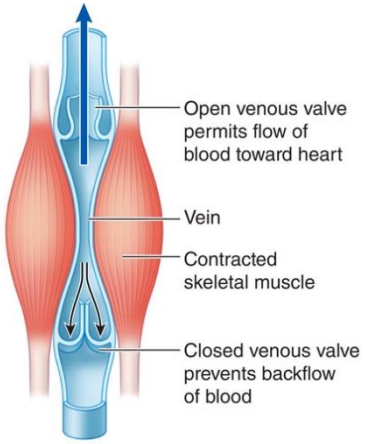
Venous return

- Five additional **extrinsic factors** increasing venous return
  - **Sympathetic activity** causes vasoconstriction
  - **Skeletal muscle pump** compresses the vein during muscle contraction
  - **Venous valves** prevent backflow of blood
  - **Respiratory activity** lowers pressure in thoracic cavity, drawing venous blood toward heart
  - **Cardiac suction** lowers pressure within heart during relaxation





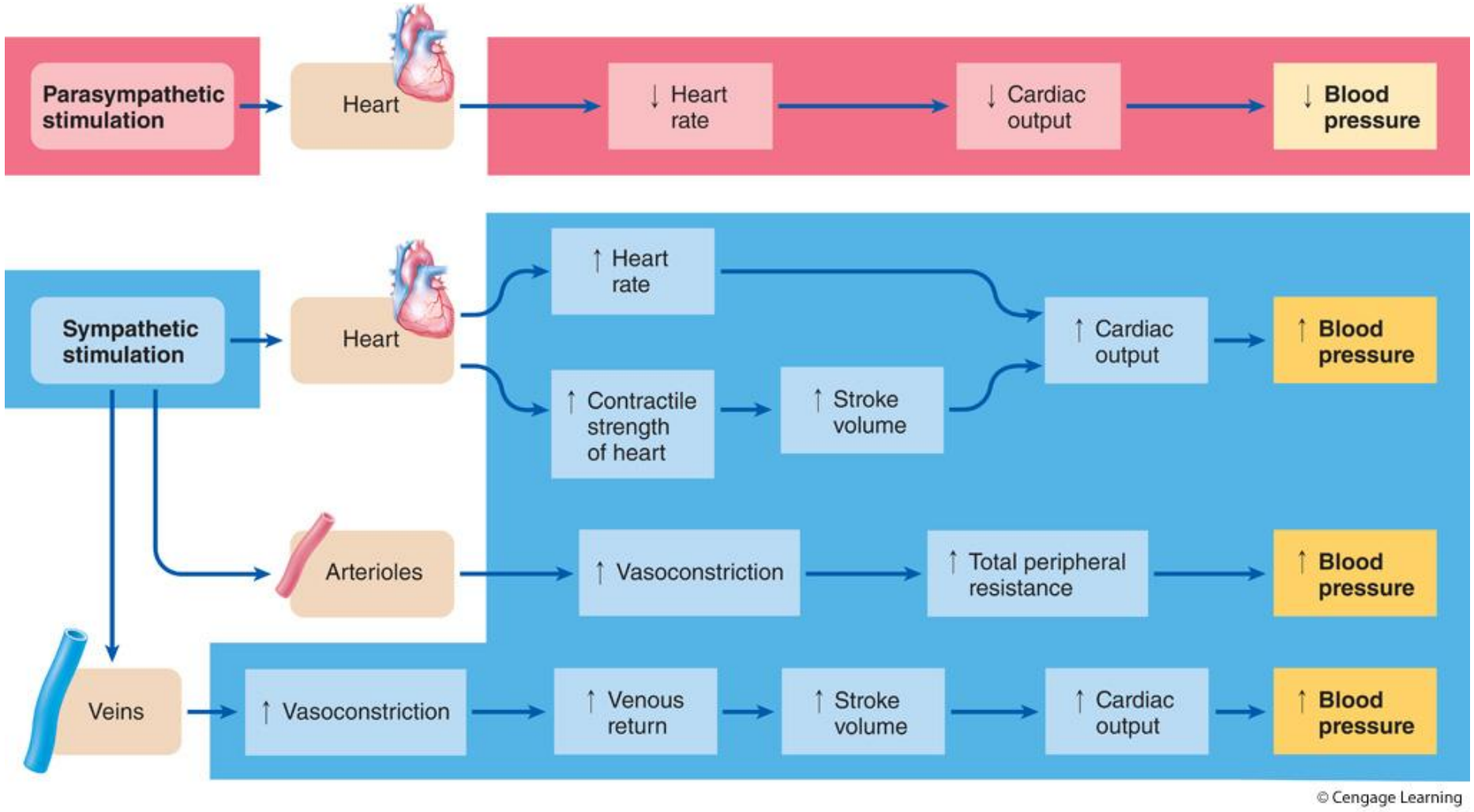
(a) Fluid moving in both directions on squeezing a fluid-filled tube



(b) Action of venous valves, permitting flow of blood toward heart and preventing backflow of blood

10.6 Blood Pressure

Mean arterial pressure

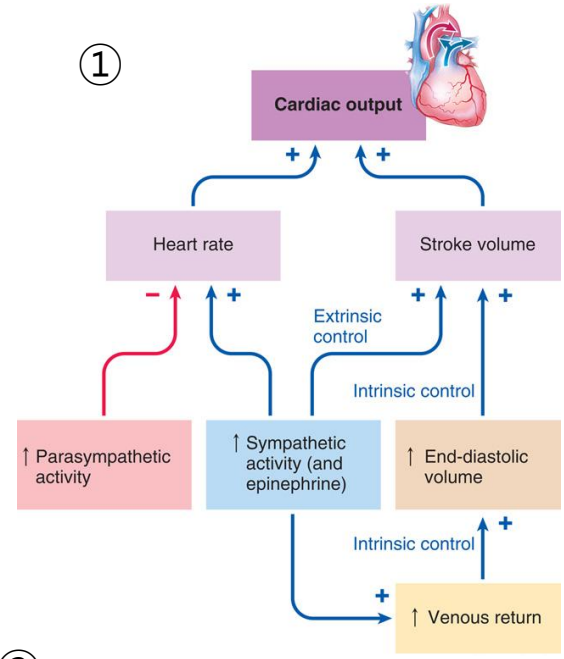


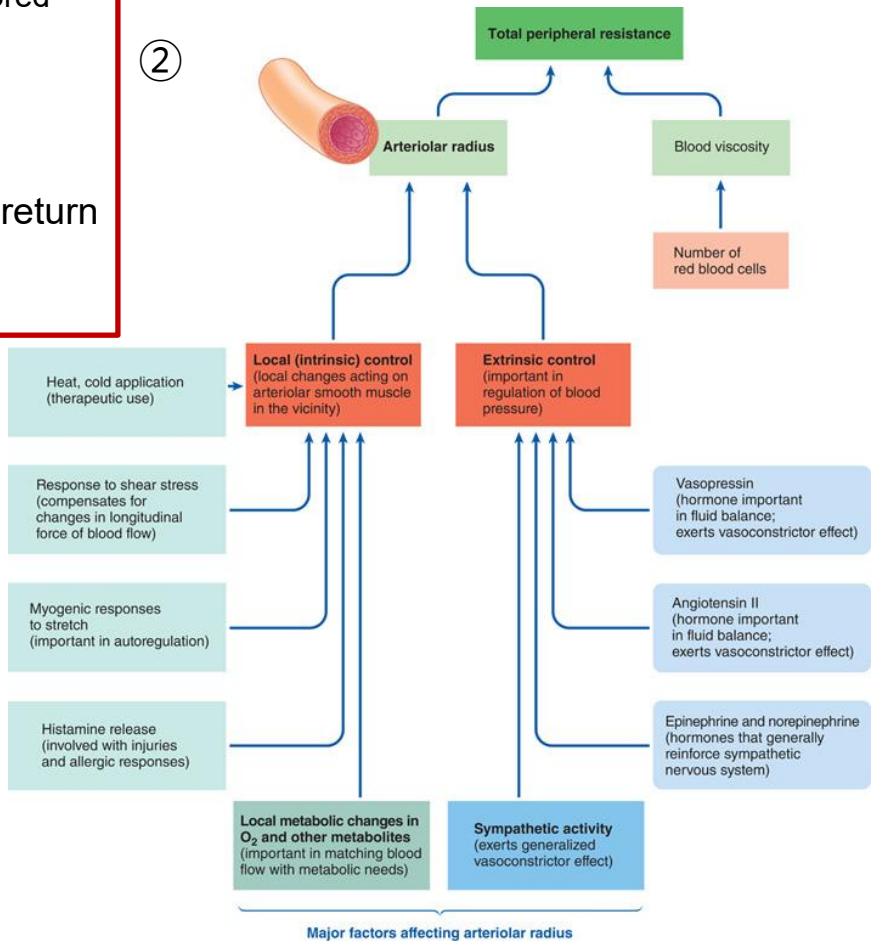
Mean arterial pressure

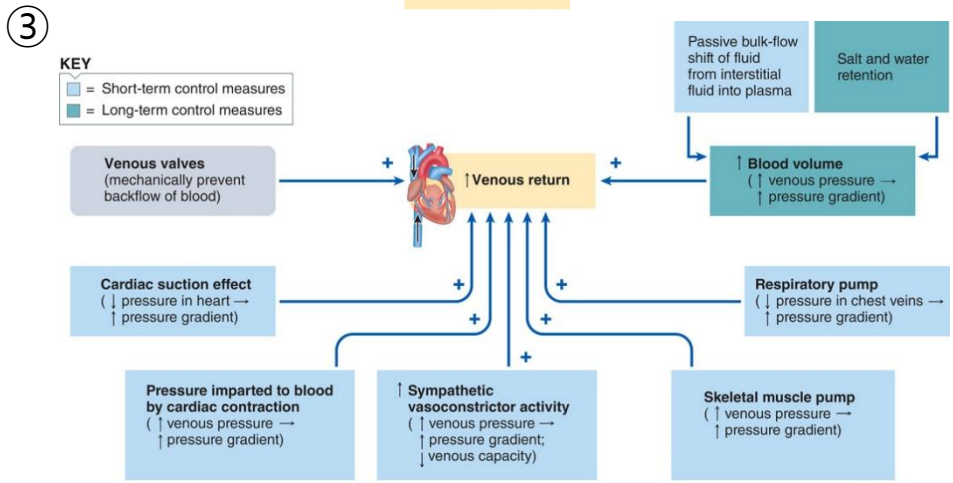
Mean arterial pressure is the blood pressure monitored and regulated in the body.

① Factors for CO
② Factors for TPR
③ Factors for venous return

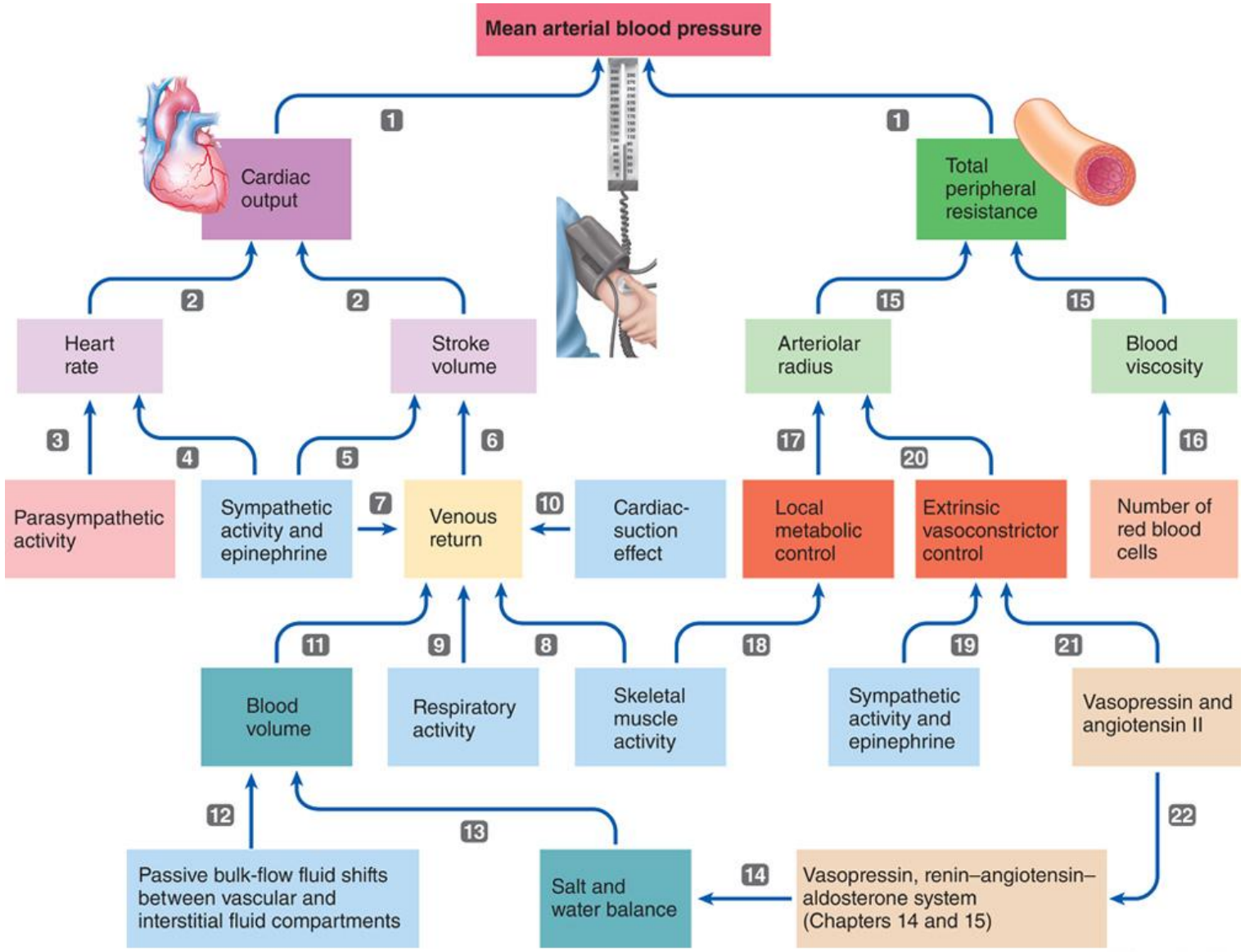
MAP = CO X TPR

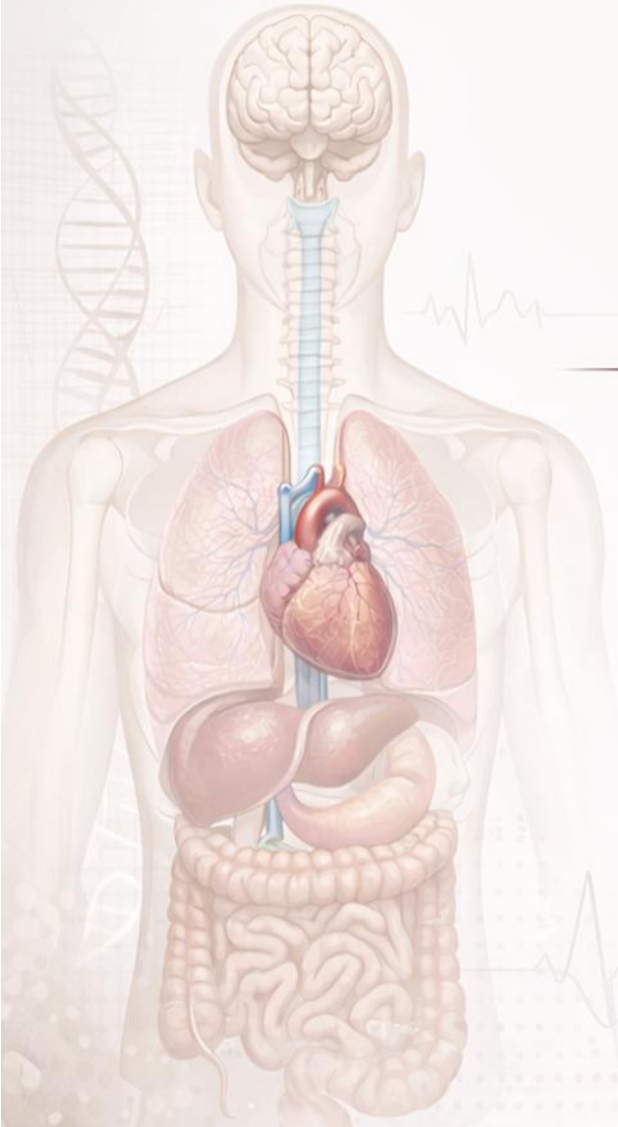






Mean arterial pressure

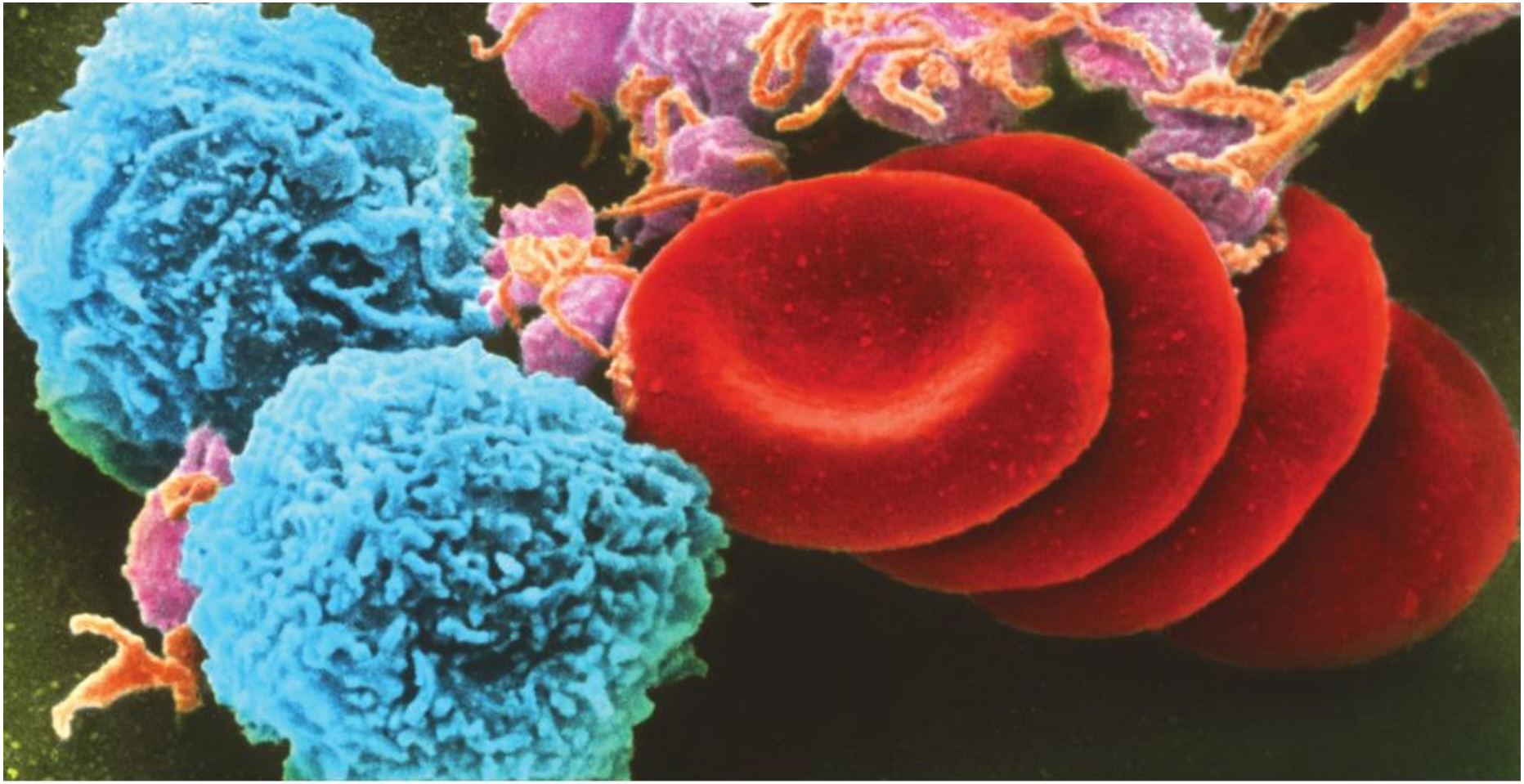




Chapter 11.
The Blood

KOREA UNIVERSITY
1905

A scanning electron micrograph of cellular elements in the blood

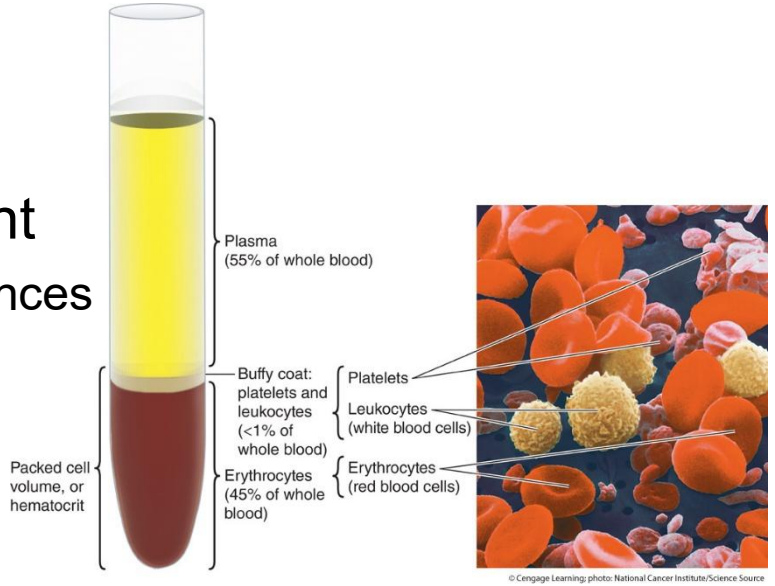


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11.1 Plasma

Plasma

- Blood represents 8% of total body weight
- Blood consists of 42-45% hematocrit (적혈구부피비; 99% of blood cells are erythrocytes), less than 1% of leukocytes plus platelets, and 55-58% plasma
- Blood consists of three cell types
  - Erythrocytes (red blood cells)
  - Leukocytes (white blood cells)
  - Platelets (thrombocytes): cell fragments
- Plasma: non-cellular liquid component
  - Consists of 90% water containing substances
    - Inorganic solutes: mostly Na^+ and Cl^-
    - Organic constituents: mostly plasma proteins
    - Other organics: waste products
      dissolved gases, hormones



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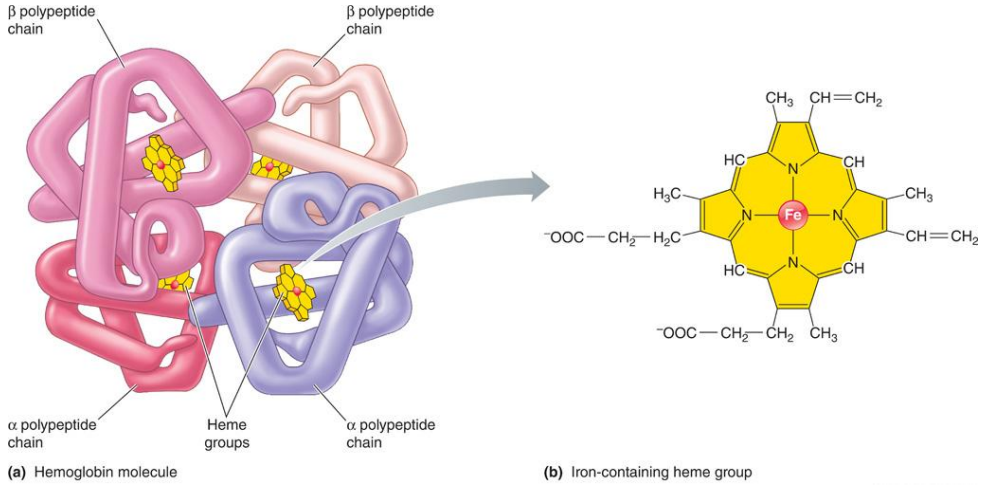
Plasma proteins

- Many of the functions of plasma are carried out by plasma proteins
  - Plasma proteins form colloid structure and do not exit to enter the interstitial fluid, to generate **osmotic gradient** that prevents loss of plasma from the capillaries into the interstitial fluid
  - Plasma proteins maintain plasma's capacity to buffer changes in pH
  - Three groups of plasma proteins perform specific tasks
    - **Albumins**: the most abundant plasma protein, colloid osmotic pressure generation + non-specific binding to poorly-soluble substances supporting their transport
    - **Globulins**: specific binding to poorly-soluble substances supporting their transport + blood-clotting + salt balance in the body
    - **Fibrinogens**: key factor in blood clotting

11.2 Erythrocytes

Erythrocytes

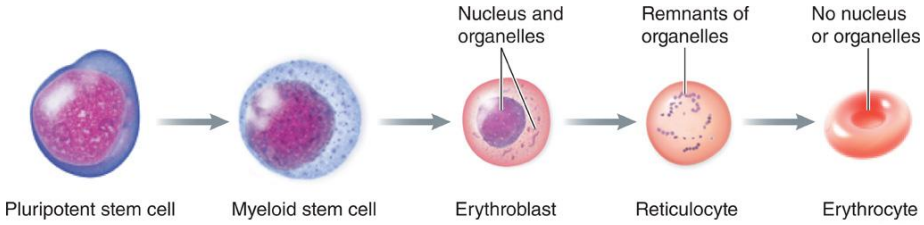
- Erythrocytes are well designed for their main function of O_2 transport in the blood
- Shape and content is ideally suited for O_2 transport
  - Flat, disc-shaped shape provide large surface area for oxygen diffusion
  - Thinness of the cell enables rapid diffusion
- Flexible membrane enables their travel through narrow capillaries
- They contain hemoglobin
  - O_2 is poorly soluble in plasma



Erythrocytes

- Hemoglobin
  - Found only in red blood cells
  - Made of globin (two α + two β polypeptide chains) and heme groups (four iron-containing groups)
  - It is a natural pigment showing red color when combined with O_2 (bluish when deoxygenated)
  - They also carry CO_2, H^+, CO, NO

- Erythrocytes lack nucleus and organelles
  - To maximize hemoglobin content

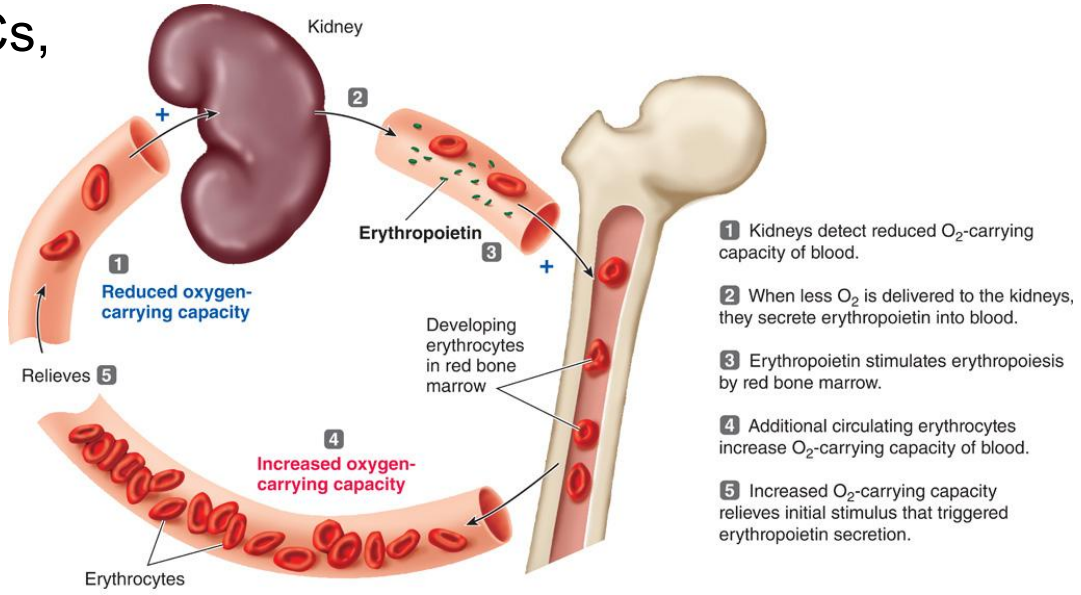


- Key erythrocyte enzymes
  - Glycolytic enzymes: cells rely on glycolysis for ATP production (no mt)
  - Carbonic anhydrase: conversion of CO_2 into bicarbonate ion ($HCO3^-$), primary form of CO_2 transport in the blood
  - CO_2 is transported by erythrocytes in two ways

Erythropoiesis

- The bone marrow continuously replaces worn-out erythrocytes
- Erythrocytes survive only 120 days (short life-span)
- Old RBCs are removed in spleen
- Erythrocytes cannot divide, and old cells must be replaced by new cells produced in the bone marrow (erythropoiesis)

- Hematopoietic stem cells in the red bone marrow produce RBCs, WBCs, and platelets
- Erythropoiesis is controlled by erythropoietin (EPO) from the kidneys



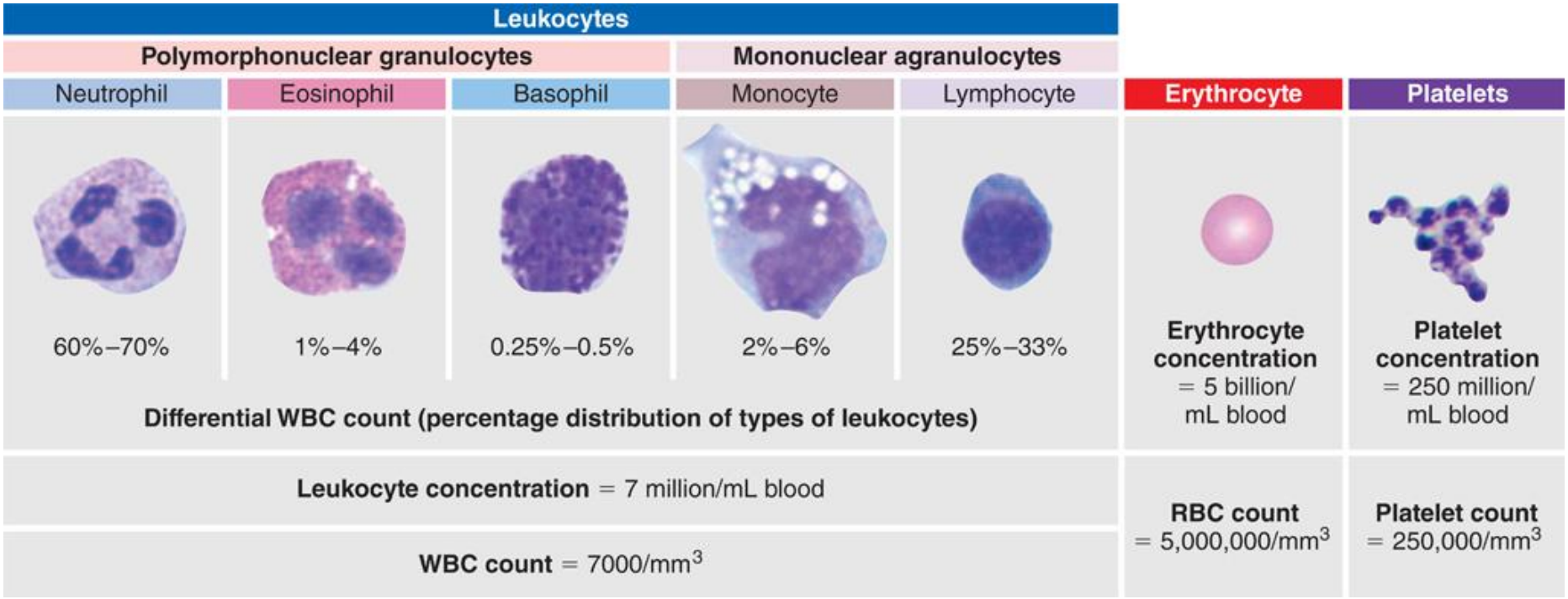
11.3 Leukocytes

Leukocytes

- Leukocytes are mobile units of the body's immune defense system
- Immunity is the body's ability to resist or eliminate potentially harmful foreign materials or abnormal cells
- Immune system: internal defense system that recognizes and either destroys or neutralizes materials that are foreign to the "normal self"

- Leukocytes primarily function as defense agents outside the blood
  - To carry out their functions, WBCs largely use a "seek out and attack", going to sites of invasion or tissue damage
  - Main reason WBCs are in the blood is for rapid transport from their site of production or storage to wherever they are needed

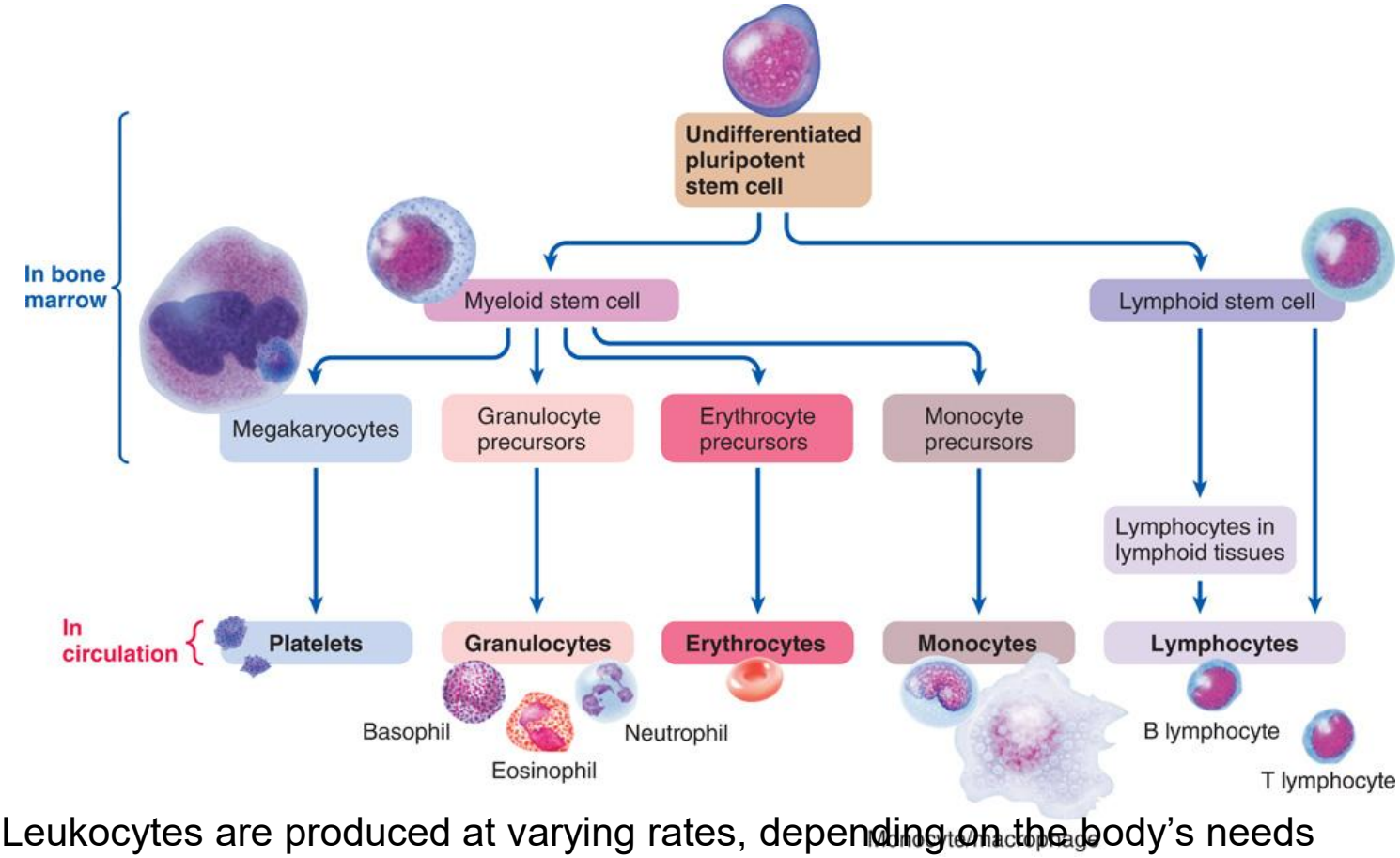
Five types of leukocytes



- Granulocytes and agranulocytes
  - Depending on appearance of their nuclei and presence or absence of granules in their cytoplasm when viewed microscopically

- Neutrophils engulf and destroy bacteria intracellularly
- Increase in eosinophils is associated with allergic conditions and parasite infestations
- Basophils synthesize and store histamine and heparin
- Monocytes perform phagocytosis
- Lymphocytes provide specific immune defense

Five types of leukocytes

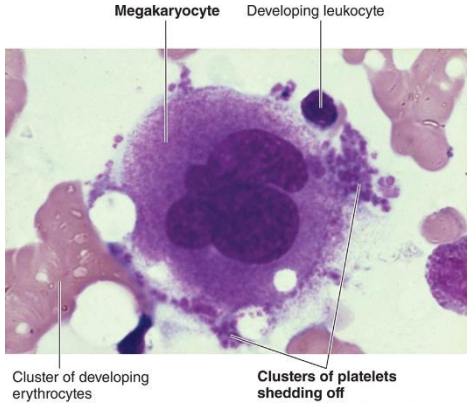


- Leukocytes are produced at varying rates, depending on the body's needs
- Abnormalities in leukocyte production (too few or too many) can be produced
- Leukemia: cancerous condition involving uncontrolled proliferation of WBC

11.4 Platelets and Hemostasis

Platelets

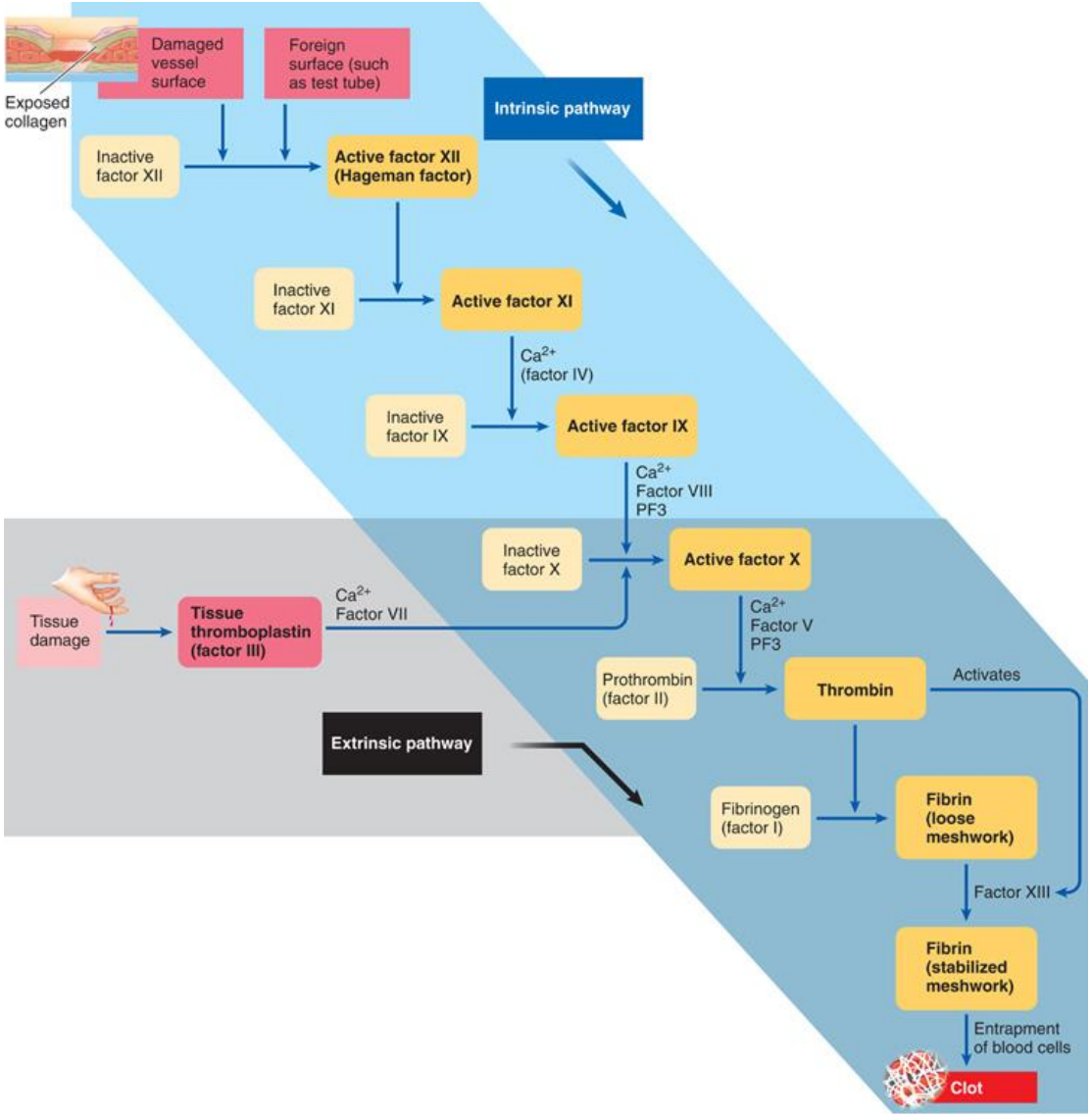
- Platelets are cell fragments shed from megakaryocytes
  - Megakaryocytes: extraordinarily large bone marrow-bound cells
  - Single megakaryocytes produce 1000 platelets
  - Thrombopoietin, hormone produced by the liver, increases the number and platelet generation of megakaryocytes
- Platelets lack nuclei, but have organelles and cytosolic enzymes essential for energy generation and protein secretion
- Platelets have actins and myosins, enabling them to contract
- Secretory and contractile activities are important for hemostasis, which prevents blood loss from damaged vessels
  - Hemostasis (지혈): Arrest of bleeding from a broken blood vessel; stopping of hemorrhage (출혈)



Hemostasis

- Three major steps of hemostasis
- Vascular spasm reduces blood flow through an injured vessel
  - Immediate constriction of injured blood vessel
- Formation of a platelet plug
  - Platelets normally do not adhere to smooth endothelium of vessels
  - Adhesion of platelets to the site of vessel injury
  - Adhered platelets become activated and release mediators such as ADP and thromboxane A_2, which recruit additional platelets
  - Aggregation of platelets then forms a platelet plug at the site of vessel injury
- Blood coagulation (clotting)
  - Transformation of blood into a solid gel
  - Conversion of fibrinogen (soluble plasma protein) into fibrin (insoluble, thread-like protein), catalyzed by thrombin
  - Fibrin adhere to damaged vessel surface and form net-like meshwork trapping blood cells including aggregating platelets

Hemostasis


Damaged vessel surface
Foreign surface (such as test tube)
Exposed collagen
Intrinsic pathway
Inactive factor XII
Active factor XII (Hageman factor)
Inactive factor XI
Active factor XI
Ca2+ (factor IV)
Inactive factor IX
Active factor IX
Ca2+ Factor VIII PF3
Inactive factor X
Active factor X
Tissue damage
Tissue thromboplastin (factor III)
Ca2+ Factor VII
Ca2+ Factor V PF3
Prothrombin (factor II)
Thrombin
Activates
Extrinsic pathway
Fibrinogen (factor I)
Fibrin (loose meshwork)
Factor XIII
Fibrin (stabilized meshwork)
Entrapment of blood cells
Clot